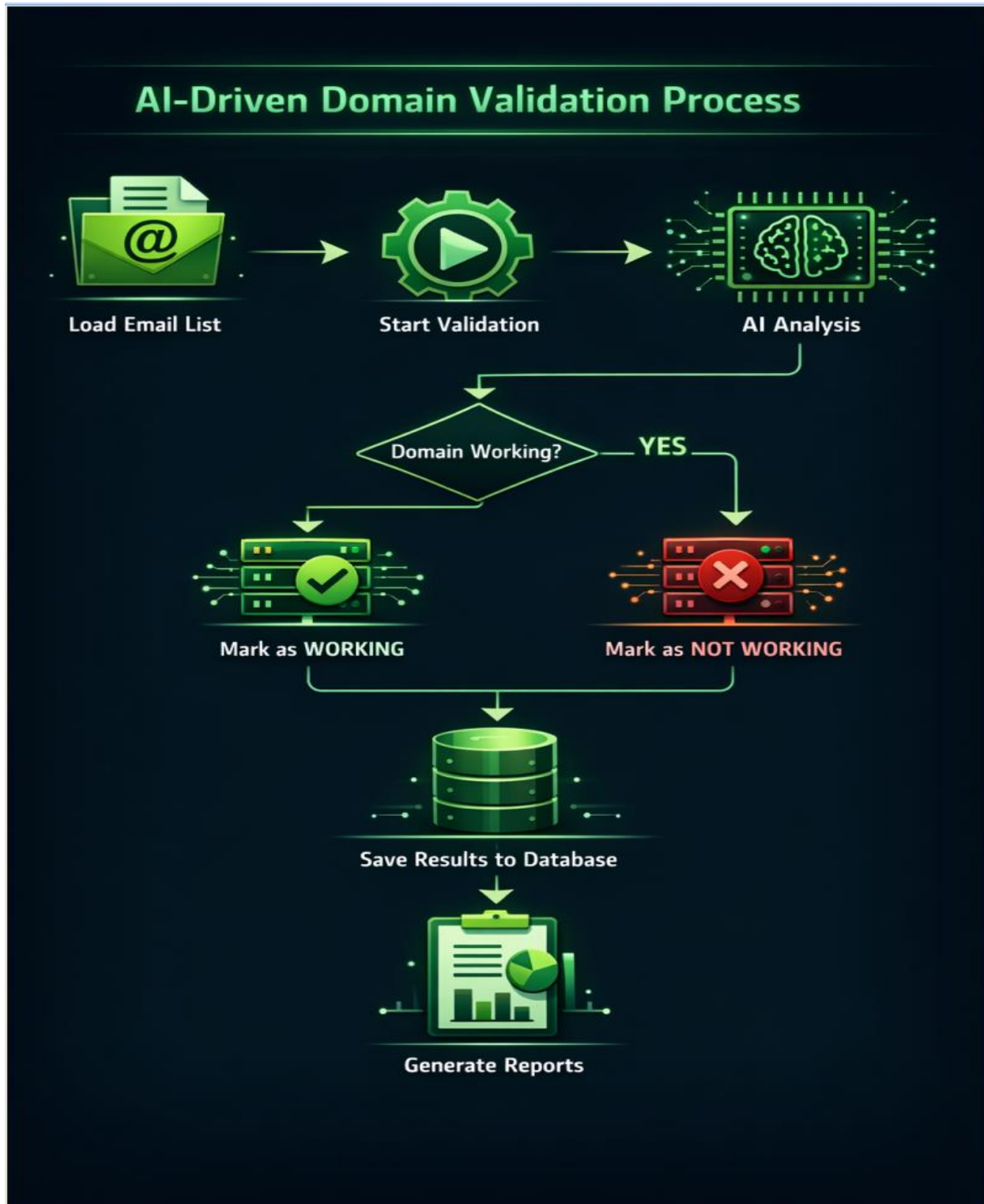
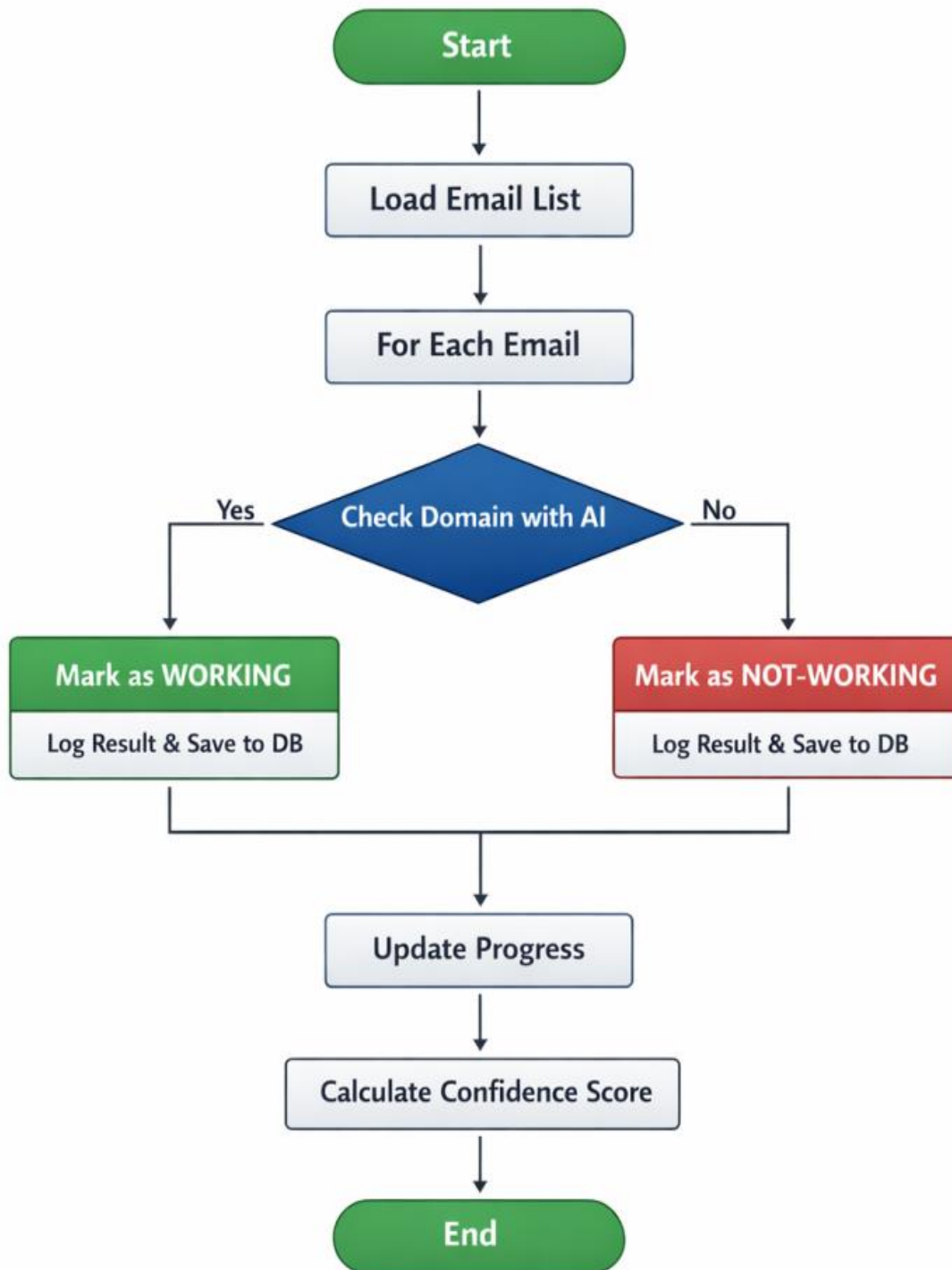


Here's a clear, industry-grade list of parameters used in domain validation logic, organized from basic syntax checks → DNS → mail infrastructure → security & reputation. This is the same logic used in email validators, signup forms, CRM systems, and anti-fraud tools.



Domain Validation Process



❑ Domain Validation Parameters (Complete Table)

Category	Parameter	What It Validates	Why It Matters
Syntax	Domain format	example.com structure	Rejects malformed domains
	Allowed characters	a-z, 0-9, hyphen	Prevents invalid domain names
	Length check	≤ 253 chars total	DNS standard compliance
	Label length	≤ 63 chars per label	DNS protocol limit
	TLD presence	.com, .org, .in	Ensures real domain
	Valid TLD	IANA approved TLDs	Blocks fake extensions
DNS	DNS resolution	Domain resolves	Domain exists
	A record	IPv4 address exists	Website/server reachable
	AAAA record	IPv6 address exists	IPv6 compatibility
	CNAME record	Alias resolution	Cloud/CDN detection
	NS records	Name servers exist	Domain is configured
Mail Infrastructure	SOA record	Start of Authority	DNS zone integrity
	MX record	Mail server exists	Domain can receive emails
	MX priority	Mail routing order	Proper mail delivery
	SMTP connection	Port 25/465/587	Mail server reachable
	SMTP banner	Server identity	Detect fake servers
	SMTP HELO/EHLO	Mail handshake	Server compliance
	SMTP response	250, 550, etc.	Mailbox/domain validity
	Catch-all detection	Accepts all emails	Avoid false positives

Security	SPF record	Sender authorization	Prevents spoofing
	SPF syntax	Valid SPF format	Mail delivery reliability
	DKIM record	Message signing	Prevents tampering
	DMARC record	Policy enforcement	Phishing protection
	TLS support	STARTTLS available	Secure mail transport
	DNSSEC	DNS integrity	Prevents DNS hijack
Reputation	Blacklist check	Spam lists	Avoid spam domains
	Greylist status	Temporary blocks	Retry logic needed
	Domain age	Creation date	New domains are risky
	WHOIS status	Active/expired	Domain legitimacy
	Registrar	Trusted provider	Risk assessment
Web Presence	HTTP response	200/301/403	Domain activity
	HTTPS support	SSL/TLS	Security trust
	SSL validity	Certificate validity	Prevent MITM attacks
	Website content	Parked/active	Business legitimacy
Geographical	IP location	Country/region	Compliance & fraud
	ASN	Hosting provider	Cloud vs shared hosting
Performance	DNS latency	Query response time	Infrastructure quality
	SMTP latency	Mail response speed	Server reliability
AI / Heuristic	Disposable domain	Temp email provider	Prevent fake signups
	Role-based email	admin@, info@	Often not personal
	Pattern anomalies	Random strings	Bot detection
	Historical behavior	Past spam activity	Risk scoring
	Confidence score	0–100%	Final decision metric
Compliance	GDPR readiness	Data handling	Legal compliance

	CAN-SPAM risk	Marketing safety	Avoid penalties
	Opt-in history	User consent	Trust validation

□ **Validation Levels (How Companies Use This)**

Level	Checks Performed	Use Case
Basic	Syntax + DNS	Signup forms
Standard	+ MX + SMTP	CRM & newsletters
Advanced	+ SPF/DKIM/DMARC	Enterprise email
AI-Driven	+ Reputation + Heuristics	Fraud prevention
Military-Grade	All parameters	Banking, Govt, FinTech

□ **Typical Validation Output**

```
{
  "domain": "example.com",
  "exists": true,
  "has_mx": true,
  "smtp_valid": true,
  "secure": true,
  "reputation_score": 92,
  "risk_level": "LOW"
}
```
